

HE2003 Econometrics I

Tutorial 1 (Week 2) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Lecture coverage:** Lecture 1 — *Introduction: why we study econometrics, correlation vs. causality, the linear regression model previewed via the California class-size example, Type I/Type II errors, and randomised controlled trials.*
- **Questions:** 4 questions from Tutorial Question Set No. 1.
- **Notation:** as in lectures — β_0, β_1 population parameters; u_i population error term (never $\hat{u}_i$ at this stage, since no data has been fitted yet); $E[\cdot]$ population expectation; $E[\cdot | \cdot]$ conditional expectation.

Question 1 (Correlation vs. causality: the three cases)

Problem

Simply observing a correlation between two variables X and Y is not enough to conclude that X has a causal effect on Y . When X and Y are correlated, there are several possible cases. Explain these cases with specific examples. (*Tutorial Question Set No. 1, Q1.*)

Solution

Method 1: case enumeration

- (i) **Forward causality** $X \rightarrow Y$: wind $\rightarrow$ windmill rotation; $\text{windmill}_i = \beta_0 + \beta_1 \text{wind}_i + u_i$ identifies the causal slope.
- (ii) **Reverse causality** $Y \rightarrow X$: crime $\rightarrow$ police staffing (cities hire police when crime rises); regressing crime on police gives a spurious positive slope.
- (iii) **Common factor** $X \leftarrow C \rightarrow Y$, no arrow between X, Y : temperature $\rightarrow$ {ice-cream sales, drowning deaths}; COVID $\rightarrow$ {mask use, online-working hours}.

3 cases when X, Y are correlated: (i) $X \rightarrow Y$, (ii) $Y \rightarrow X$, (iii) $C \rightarrow X, C \rightarrow Y$.

Question 2 (The class-size / test-score linear regression model)

Problem

Explain the linear regression model in the example of class size and student performance. What are the parameters of the model? How to interpret these parameters? What is the meaning of the error term? (*Tutorial Question Set No. 1, Q2.*)

Solution

Method 1: model and parameters

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i, \quad i = 1, \dots, n \quad (n = 420 \text{ districts})$$

- TestScore_i : average test score of district i (dependent variable); STR_i : students per teacher (regressor).
- β_0 (intercept): value of the population line at $\text{STR} = 0$; it only anchors the line.
- $\beta_1 = \Delta \text{TestScore} / \Delta \text{STR}$ (slope): points of test score per one-student-per-teacher rise in STR ; e.g. $\beta_1 = -2$, $\Delta \text{STR} = -2 \implies \Delta \text{TestScore} = (-2)(-2) = 4$ points.
- u_i : all other factors affecting district i 's score (income, parental education, teacher quality, ...).

$$\begin{aligned} \text{TestScore}_i &= \beta_0 + \beta_1 \text{STR}_i + u_i, & \beta_1 &= \Delta \text{TestScore} / \Delta \text{STR}, \\ u_i &= \text{all other factors affecting TestScore}_i. \end{aligned}$$

Question 3 (Conditional and unconditional moments of $Y = X^2 + Z$)

Problem

Let X and Z be two independently distributed standard normal random variables, and let $Y = X^2 + Z$.

- (a) Show that $E[Y | X] = X^2$.
- (b) Show that $E[Y] = 1$.
- (c) Show that $E[XY] = 0$.

(Tutorial Question Set No. 1, Q3.)

Solution

Facts.

$$\begin{aligned} E[X] = E[Z] = 0, \quad E[X^2] &= \text{Var}(X) + E[X]^2 = 1 \\ E[X^3] &= 0 && \text{(odd moment)} \\ X \perp Z &\implies E[XZ] = E[X]E[Z], \quad E[Z | X] = E[Z] \end{aligned}$$

Method 1: direct computation

(a)

$$\begin{aligned} E[Y | X] &= E[X^2 | X] + E[Z | X] && \text{(linearity)} \\ &= X^2 + E[Z] && (X^2 \text{ is a function of } X; X \perp Z) \\ &\implies \boxed{E[Y | X] = X^2.} \end{aligned}$$

(b)

$$\begin{aligned} E[Y] &= E[X^2] + E[Z] = 1 + 0 && \text{(linearity)} \\ &\implies \boxed{E[Y] = 1.} \end{aligned}$$

(c)

$$\begin{aligned} E[XY] &= E[X^3] + E[XZ] && \text{(expand)} \\ &= 0 + E[X]E[Z] = 0 && (X \perp Z) \\ &\implies \boxed{E[XY] = 0.} \end{aligned}$$

Method 2: law of iterated expectations

- (b) $E[Y] = E[E[Y | X]] = E[X^2] = 1$ (LIE, (a))
- (c) $E[XY] = E[X E[Y | X]] = E[X^3] = 0$ (LIE, pull out X , (a))

Correlation. $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - 0 \cdot 1 = 0 \implies \text{Corr}(X, Y) = 0$,
yet $E[Y | X] = X^2$ varies with X .

$\begin{array}{l} E[Y X] = X^2, \quad E[Y] = 1, \quad E[XY] = 0, \\ \text{yet } \text{Corr}(X, Y) = 0 \text{ while } Y \text{ depends strongly on } X. \end{array}$

Question 4 (Type I and Type II errors and the trade-off)

Problem

Explain the Type I and Type II errors and the trade-off between the two errors by using a specific example. (*Tutorial Question Set No. 1, Q4.*)

Solution

Method 1: COVID ART-test example

H_0 : healthy, H_1 : sick; reject H_0 iff antibody level $> c$, with $c = 10$.

- **Type I** (false positive): reject H_0 when H_0 is true; probability α (the significance level).
- **Type II** (false negative): fail to reject H_0 when H_1 is true; probability β .
- **Trade-off**: $c : 10 \rightarrow 100$ gives $\alpha \downarrow$, $\beta \uparrow$ (infected with levels in $(10, 100)$ missed);
 $c : 10 \rightarrow 1$ gives $\beta \downarrow$, $\alpha \uparrow$ (healthy with levels in $(1, 10)$ flagged).

Method 2: tail areas of the densities

Antibody level L : density, CDF f_0, F_0 under H_0 ; f_1, F_1 under H_1 ; $f_0(c), f_1(c) > 0$:

$$\begin{aligned}\alpha &= P(L > c | H_0) = 1 - F_0(c) & \implies \alpha'(c) = -f_0(c) < 0 \\ \beta &= P(L \leq c | H_1) = F_1(c) & \implies \beta'(c) = f_1(c) > 0\end{aligned}$$

$\therefore c \uparrow \implies \alpha \downarrow, \beta \uparrow$; the densities overlap ($f_0, f_1 > 0$ on a common range), so no c makes both zero.

Type I error = $P(\text{reject } H_0 H_0 \text{ true}) = \alpha$, Type II error = $P(\text{fail to reject } H_0 H_1 \text{ true}) = \beta$, $\alpha \downarrow \implies \beta \uparrow$ (and vice versa).

Final Answers

1. Three cases when X, Y are correlated: (i) $X \rightarrow Y$ (e.g. wind $\rightarrow$ windmill rotation); (ii) $Y \rightarrow X$ (reverse causality, e.g. crime $\rightarrow$ police staffing); (iii) common factor $C \rightarrow X, C \rightarrow Y$ (e.g. temperature $\rightarrow$ {ice-cream sales, drowning deaths}; COVID $\rightarrow$ {mask use, online-working hours}).
2. $\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i$; β_0 is the intercept (value at $\text{STR} = 0$); $\beta_1 = \Delta \text{TestScore} / \Delta \text{STR}$ (points per unit STR; e.g. $\beta_1 = -2 \Rightarrow \Delta \text{TestScore} = 4$ for $\Delta \text{STR} = -2$); u_i is all other factors affecting test score.
3. $E[Y | X] = X^2$; $E[Y] = 1$; $E[XY] = 0$; yet $\text{Corr}(X, Y) = 0$ while Y depends strongly (nonlinearly) on X .
4. Type I error = $P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$; Type II error = $P(\text{fail to reject } H_0 \mid H_1 \text{ true}) = \beta$. Raising the ART threshold ($10 \rightarrow 100$) lowers α but raises β ; lowering it ($10 \rightarrow 1$) reverses this.